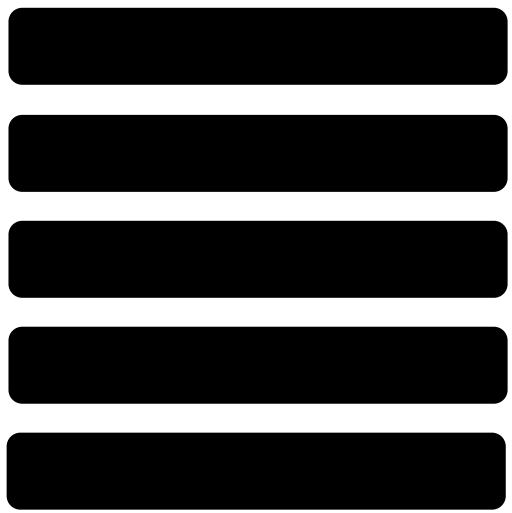


Row-Stores vs. Col-Stores

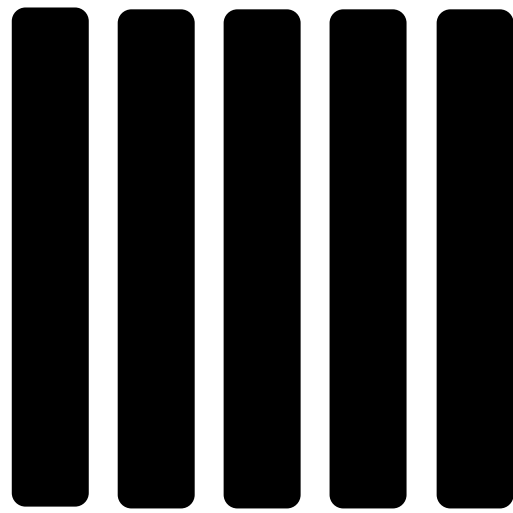
Relational Databases are Everywhere



row-stores



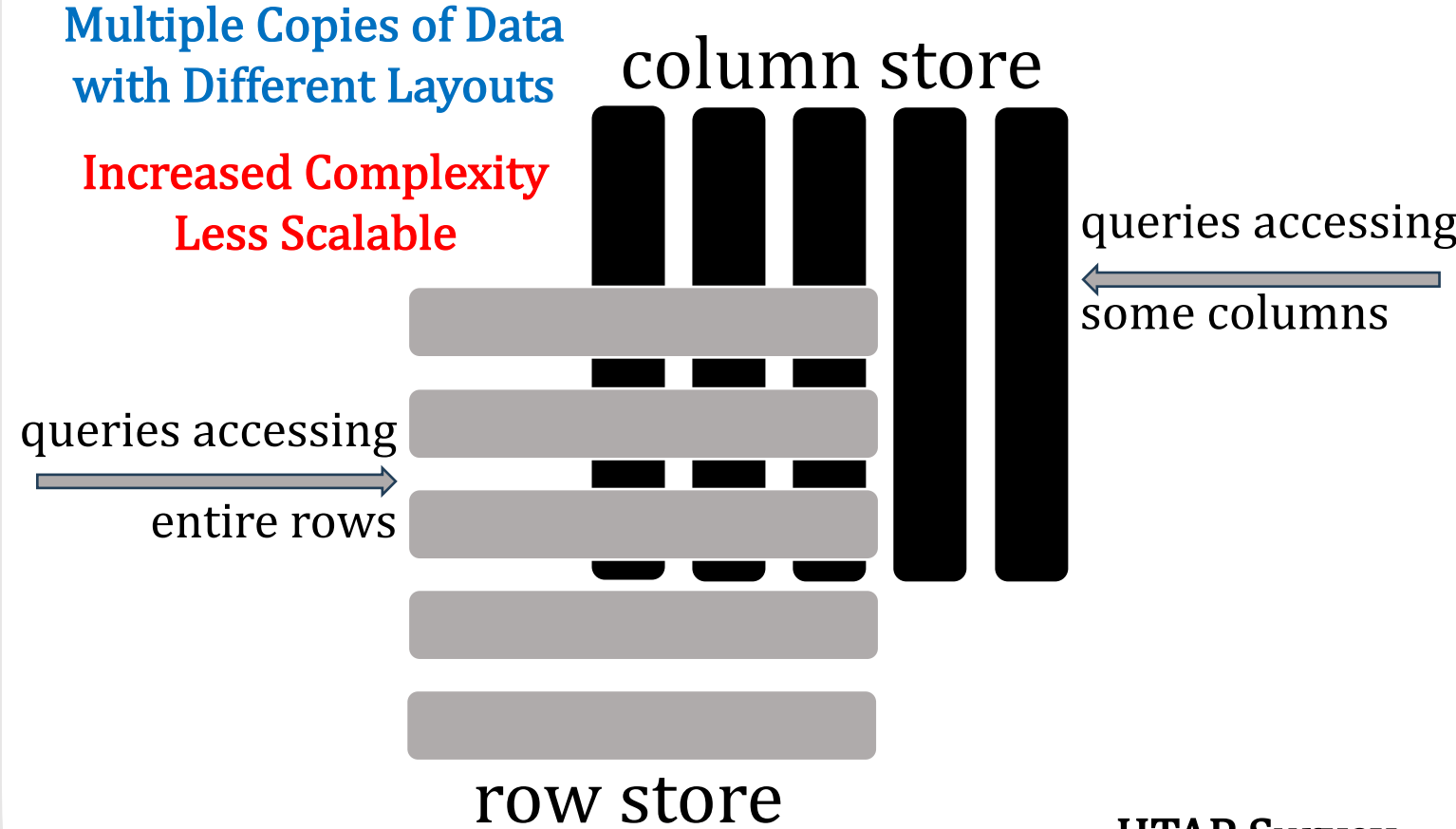
column-stores



Hybrid and Adaptive Layouts

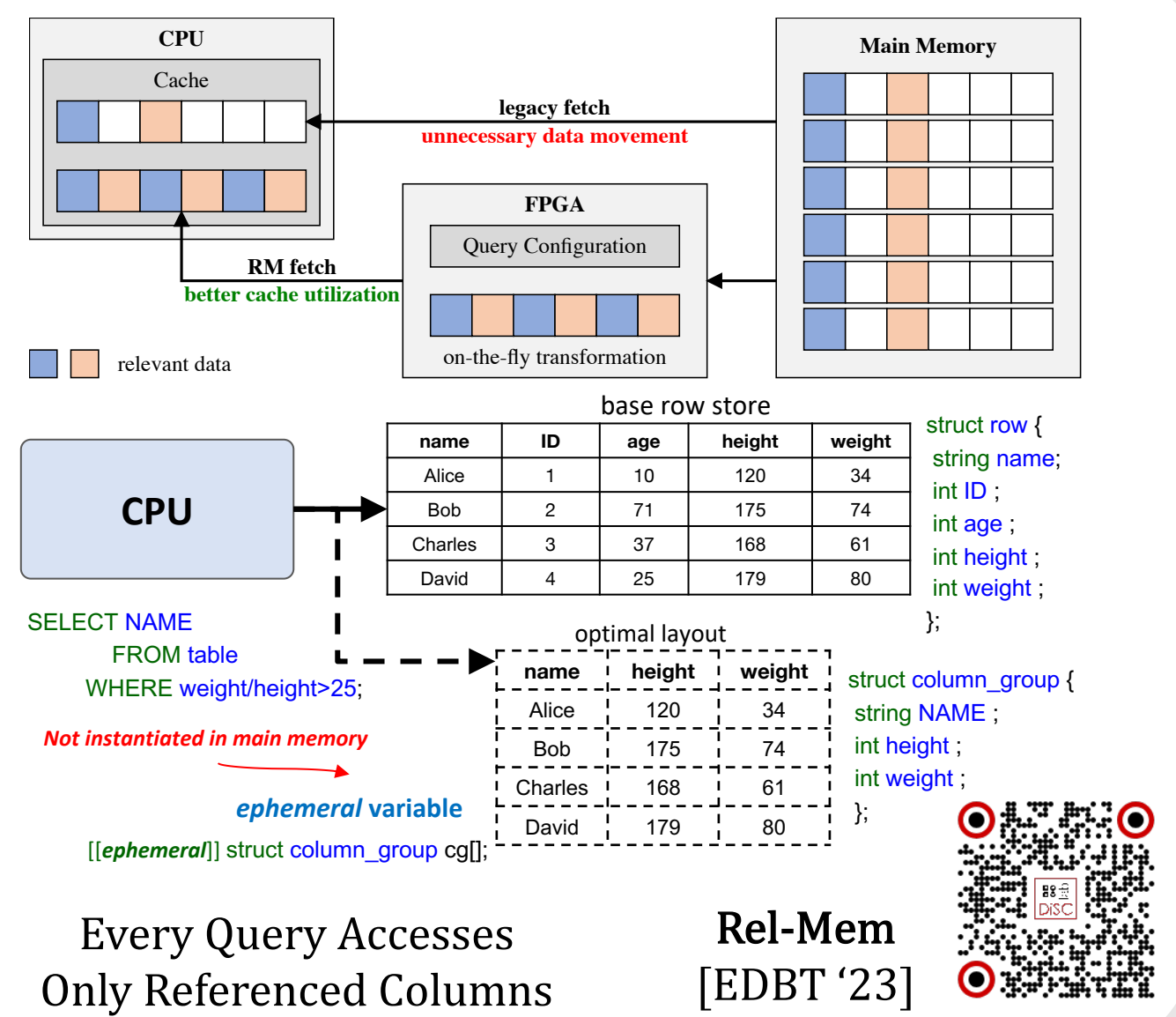
Multiple Copies of Data
with Different Layouts

Increased Complexity
Less Scalable



HTAP Survey
[SIGMOD '17]

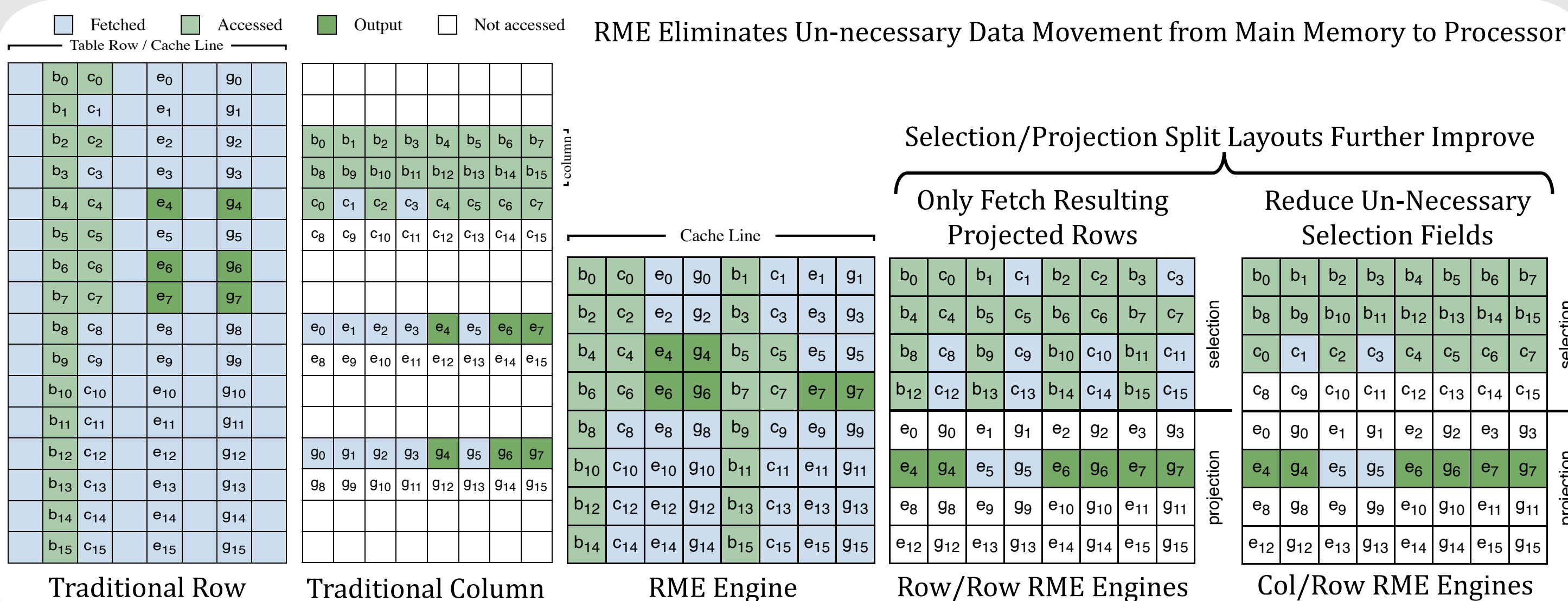
Relational Memory Engine



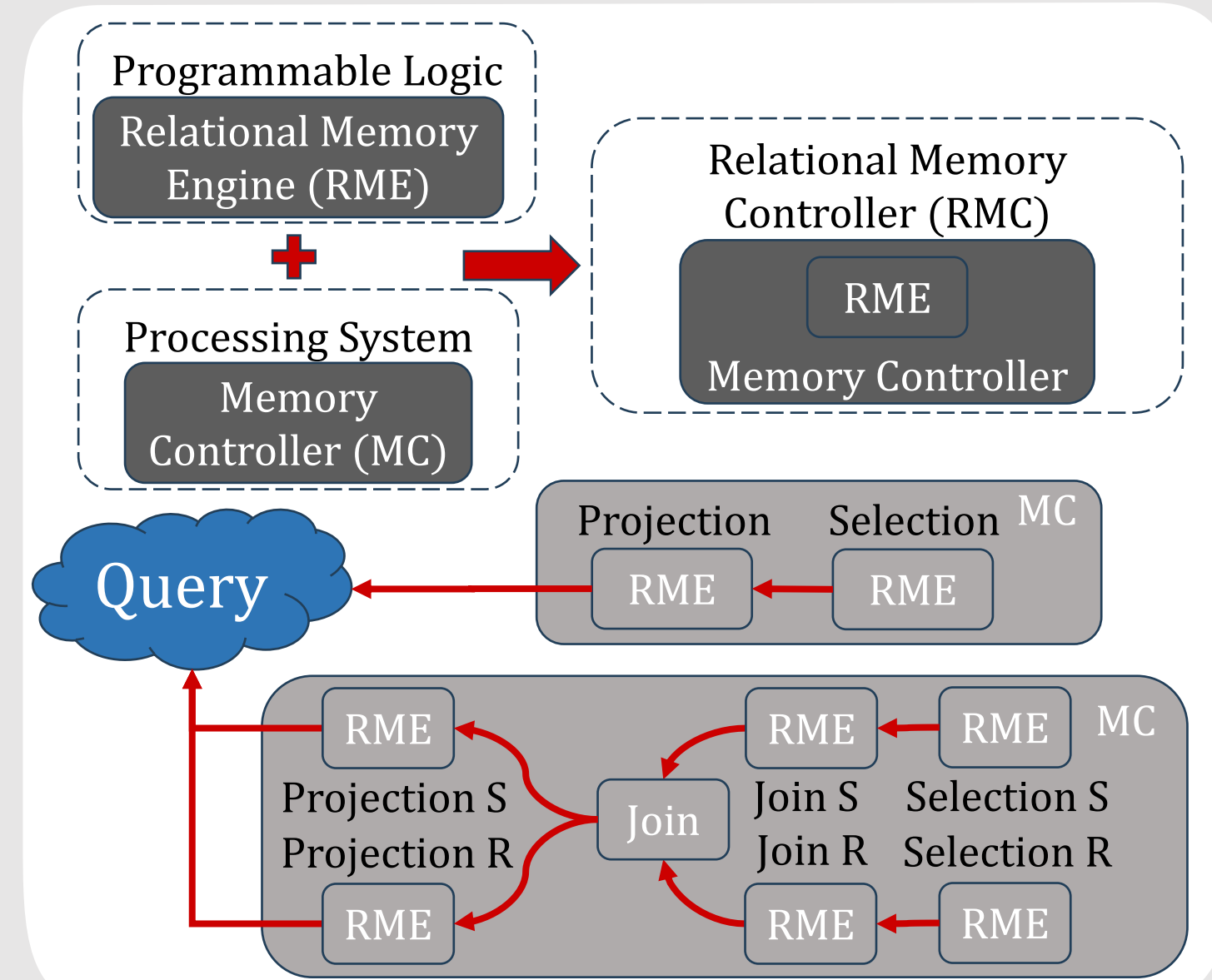
Every Query Accesses
Only Referenced Columns

Rel-Mem
[EDBT '23]

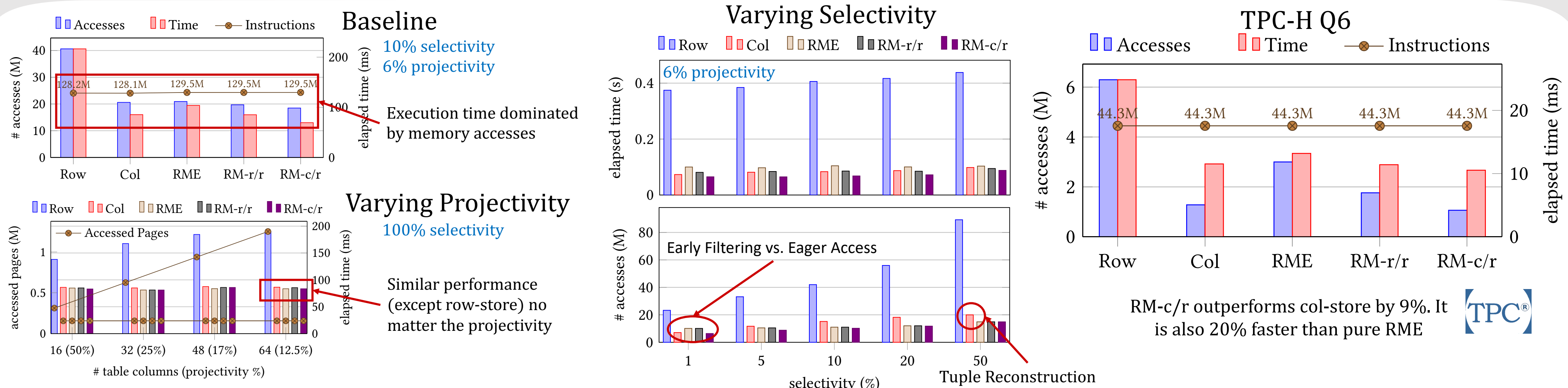
Split Layouts for Selection and Projection with Relational Memory



Broader Vision



Preliminary Results

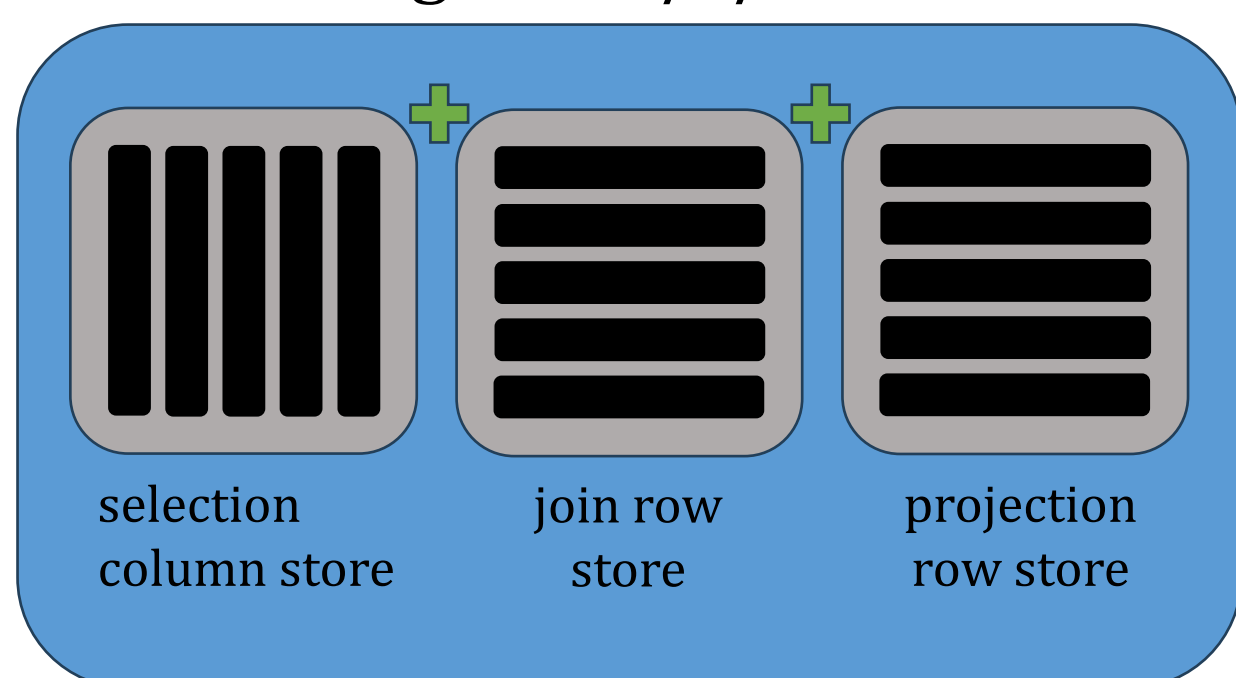


Split Layouts allow for much smaller memory footprint than ROW store and comparable performance to COL store

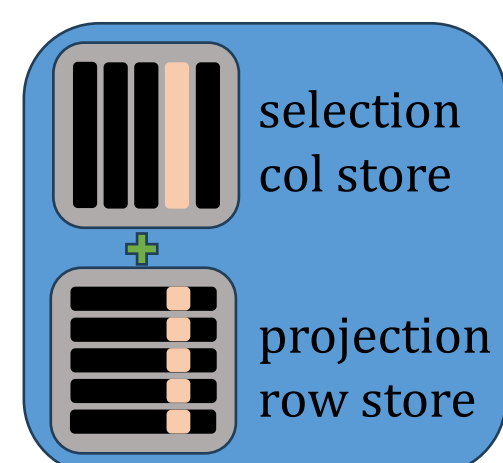
Next Steps

Handling Joins

e.g. RM-c/r/r



Overlapping Selection/Projection



Data vs Tuple
Duplication Reconstruction

Emulation in hardware for faster development, deployment, and testing, in a tighter integration with:



Conclusion

One RME already offers near-optimal layouts

Split Layouts use multiple RMEs to offer different layouts for selection/projection

Transparent to the user

RME C/R minimizes memory accesses and time

Up to 1.55x faster than RME due to 65% less cache transfers

